JYOTIRMAY SRIVASTAVA

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Kolkata, West Bengal, India.

EDUCATION

MS by Research - IISER Thiruvananthapuram Kerala, India Aug 2020 ▸ July 2024
3 year Program + 1 year Research
School of Biology
CGPA: 8.6/10
Research Focus: Neuroscience
Courses: Molecular Biology, Neuroscience, Animal Physiology, Structural Biology, Genetics, etc.
Research: Parkinson's Disease research in Dr. Poonam Thakur's Lab (January 2022 - June 2024)

BSc (Hons) - Chaudhary Charan Singh University, Meerut Uttar Pradesh, India Aug 2017 ▸ July 2020
School of Life Sciences
Major in Microbiology | Percentage: 71.35
Courses: Molecular Biology, Virology, Mycology, Bacteriology, etc.

PUBLICATIONS AND CONFERENCES

Kachappilly, N., **Srivastava, J.**, Swain, B. P., & Thakur, P. (2022). **Interaction of alpha-synuclein with lipids.** In *Methods in Cell Biology* (Vol. 169, pp. 43-66). Academic Press.

Srivastava, J. & Ray, S. (2026). **Multisensory integration for navigational memory in Periplaneta americana** In *No Garland Neuroscience, IISER Mohali & BRIC-NABI, 2026*

SKILLS

- IHC/ICC, Microscopy (Confocal)
- Mouse handling (maintenance, Intraperitoneal (IP) / Subcutaneous (SC) injections)
- Behavioural experiments (mouse, insects)
- Stereotactic Surgery, Brain dissection
- Basic Electronics (working with Arduino)
- DeepLabCut, SLEAP, Argos, Bonsai
- Programming (Python, R, Bash)
- Cloning, PCR, Protein purification (SEC, Ion Exchange, Ni-NTA)
- Mammalian cell culture, Western blotting
- Insect handling, dissection

AWARDS AND HONORS

CSIR **National Eligibility Test** for Junior Research Fellowship : 68th Rank, Percentile- 99.69 (All-India) 2025
Parkinson's Foundation Visiting Research Fellowship 2024
Department of Biotechnology -Junior Research Fellowship 2023
Indian Institute of Technology (IIT) **Joint Admission Test for Masters** (JAM) : 57th Rank (All-India) 2020
TIFR **Nationwide Entrance Examination for Biology** (JGEEBILS) 2020

PROJECTS AND EXPERIENCE

Research Associate: Dr Subhasis Ray at CHINTA, TCG CREST Aug 2025 ▸ Now
Understanding the neural mechanisms of multisensory integration in Cockroaches

- Designing behavioural experiments to constrain and test sensory manipulations and navigational memory
- Designing apparatus for simultaneous presentation of stimuli of multiple sensory modalities, with plans for concurrent live electrode recordings from the insect brain

Skills Gained: Insect handling, behavioural experiments, electronics

Master's Thesis: Evaluation of the neuroprotective potential of 6-BIO treatment in a chronic mouse model of PD April 2022 ▸ April 2024
Evaluating a potential Parkinson's Disease treatment avenue

- Developing a more physiologically relevant mouse model of Parkinson's Disease
- Performing stereotactic injections in mouse brain to target substantia nigra, regular follow-ups for behavioural assessment to estimate neurodegeneration
- Studying the changes in key molecular players involved in PD pathogenesis and mechanism of action of the drug

Skills Gained: Mouse handling, survival surgeries, immunohistochemistry, confocal microscopy, western blotting

Remote Internship: Dr. Sukant Khurana at CSIR-CDRI Jul 2019 ▸ Dec 2019
Learnt bioinformatics and RNAseq data analysis

- Got introduced to the pathology of cancer and the role of various ion channels, especially HCN ion channels, in the progression of the disease
- Learnt RNA-seq data analysis, using live and novel RNA-seq datasets generated from wet-lab experiments in the lab
- Learnt the intricacies of taking raw-data and extracting useful insights from the same to guide further wet-lab experiments

Skills Gained: RNA-seq data analysis, Python, R

Summer internship: Dr. Sanjay at Credora Life Sciences Jul 2019 ▸ Oct 2019
Learnt mammalian cell culture and q-PCR

- Got introduced to mammalian cell culture, and basic cell-based techniques (immunocytochemistry, and MTT cell-viability assay)
- Learnt basic techniques in molecular biology (DNA/RNA isolation, PCR, and q-PCR)
- Designed and executed a short project to identify and check the efficacy of a new anti-neoplastic drug combinations

Skills Gained: Cell culture, molecular biology techniques

CERTIFICATES

NeuroAI Neuromatch Academy 2024

Computational Neuroscience Neuromatch Academy 2022

Genomic Data Science Specialization John's Hopkins University (COURSERA) 2021

Bioinformatics I&II Toronto University (COURSERA) 2019

Medical Neuroscience Duke University (COURSERA) 2019

LANGUAGES

English IELTS Band 8.0 (2026) **Hindi** Native

MEMBERSHIPS

Indian Academy of Neuroscience Permanent Member 2022